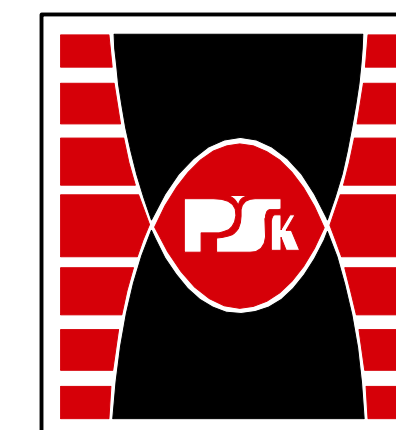




RANDOM-FOREST MODELLING OF LOCAL HEAT TRANSFER COEFFICIENTS DURING FLOW BOILING IN MINICHANNELS: A REPRODUCIBLE PYTHON PIPELINE AND TIME-BLOCKED VALIDATION

ABSTRACT/INTRODUCTION

A reproducible Python-based workflow was developed for estimating and modelling local heat transfer coefficients during flow boiling in minichannels. Two experimental datasets were analysed: FC-72 in a horizontal multi-channel module and FC-770 in a vertical few-channel module. Local HTC was calculated from Newton's law using a regime-dependent reference temperature. A ± 2 K temperature-difference deadband was applied to distinguish subcooled, ambiguous and saturated samples. Random-Forest regression was trained in log-space and assessed using time-blocked validation consistent with the transient character of the experiments.

AIM OF THE STUDY

The aim of the study was to develop and test a reproducible Python-based workflow for local HTC reconstruction and Random-Forest modelling in minichannel flow-boiling experiments. The workflow harmonises two datasets, applies a regime-dependent HTC definition, uses a ± 2 K deadband for regime classification, and compares model behaviour separately for subcooled and saturated regimes.



Figure 1.
View of the
experimental
setup

EXPERIMENTAL DATASETS

Two experimental datasets obtained at Kielce University of Technology were analysed.

Series 1: FC-72. The first dataset was obtained in a horizontally oriented module with 17 parallel rectangular minichannels. After preprocessing and spatial coverage harmonisation, the ML-ready dataset contained 536 time frames and 4288 frame-by-bin samples.

Series 2: FC-770. The second dataset was obtained in a vertical module with five parallel rectangular minichannels. After preprocessing, the ML-ready dataset contained 385 time frames and 3850 frame-by-bin samples.

In both series, the minichannels were asymmetrically heated by a thin Haynes-230 alloy foil. Wall temperature fields were measured by infrared thermography, while inlet/outlet fluid temperatures, pressure, mass flow rate and electrical parameters were recorded to reconstruct local thermal and hydraulic quantities.

EXPERIMENTAL SETUP

The experiments were carried out on a closed-loop flow-boiling stand equipped with a test section, circulation pump, pressure-control tank, auxiliary heat exchanger, Coriolis mass flow meter, pressure transducers, thermocouples and electrical power supply (Figure 1). Infrared thermography was used to measure the heated-wall temperature distribution. High-speed visualisation provided qualitative regime context only.

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DATA PROCESSING

Raw experimental signals were processed in Python and converted into local frame-by-bin records. The workflow included reconstruction of absolute pressure, calculation of pressure-based saturation temperature, interpolation of bulk fluid temperature, correction of heat flux for external losses, axial binning of infrared wall-temperature data and calculation of local temperature differences.

Local HTC was calculated from Newton's law:

$$h = q / (T_w - T_{f,ref}) \quad (1)$$

where q is the corrected heat flux, T_w is the wall temperature measured by infrared thermography, and $T_{f,ref}$ is the regime-dependent reference temperature: $T_{f,ref} = T_{f,bulk}$ for subcooled boiling and $T_{f,ref} = T_{sat}$ for saturated boiling.

The wall-to-saturation temperature difference was defined as $\Delta T_{w,sat} = T_w - T_{sat}$.

Samples were classified as subcooled for $\Delta T_{w,sat} < -2$ K, saturated for $\Delta T_{w,sat} > +2$ K, and ambiguous for $|\Delta T_{w,sat}| \leq 2$ K.

ML METHOD/VALIDATION

A Random-Forest regressor was used as a baseline model for local HTC prediction. The target was transformed to $\log_{10}(\text{HTC})$. Input features included operating conditions and local temperature-difference variables. Performance was evaluated using time-blocked validation, with consecutive frame ranges used as test segments.

RESULTS

Random-Forest regression showed better and more stable performance in the saturated regime than in the subcooled regime. Across stable time blocks, saturated-regime predictions achieved high accuracy for both datasets, with $R^2(\log_{10}(\text{HTC}))$ close to 0.97 (Table 1). Subcooled-regime predictions were more variable and sensitive to time-block shifts. The results confirm that regime-specific reporting is necessary when applying ML models to flow-boiling data.

Table 1. Random-Forest performance under time-blocked validation. Mean $\pm$ standard deviation; errors are for $\ln(\text{HTC})$ unless stated.

Series	Regime	$R^2(\log_{10}(\text{HTC}))$ mean $\pm$ std	MAE($\ln(\text{HTC})$) mean $\pm$ std	RMSE($\ln(\text{HTC})$) mean $\pm$ std	Mean MAE% / RMSE%
Series 1 (FC-72)	Subcooled	0.634 $\pm$ 0.081	0.0427 $\pm$ 0.0163	0.0686 $\pm$ 0.0381	4.37% / 7.10%
Series 1 (FC-72)	Saturated	0.973 $\pm$ 0.0268	0.0122 $\pm$ 0.0080	0.0177 $\pm$ 0.0093	1.23% / 1.78%
Series 2 (FC-770)	Subcooled	0.544 $\pm$ 0.299	0.0673 $\pm$ 0.0279	0.0982 $\pm$ 0.0497	6.96% / 10.32%
Series 2 (FC-770)	Saturated	0.969 $\pm$ 0.0373	0.0440 $\pm$ 0.0339	0.0576 $\pm$ 0.0459	4.50% / 5.93%

DISCUSSION

Saturated boiling provided a stronger and more stable signal for log-space HTC regression than subcooled boiling. Subcooled-regime performance was more sensitive to time-block shifts and low HTC variability. Therefore, Random Forest should be treated as a reproducible HTC reconstruction and quality-control component, not as a universal heat-transfer correlation.

CONCLUSIONS

- A reproducible Python workflow for local HTC reconstruction was developed.
- Random-Forest predictions were most stable in the saturated regime.
- Regime-specific reporting is essential for reliable ML analysis of flow-boiling data.